

From Knee MRI Representations to Sparse Memory

Proposed LinkedIn project and post outline

1. Core idea

The strongest version of this project is not to claim that JEPA, BioMedCLIP, sparse autoencoders, MovieChat, and LLM memory are already one unified method. Instead, present them as a research trajectory that leads to one focused question:

Can a sparse autoencoder decompose biomedical vision-language representations into human-interpretable anatomical or pathological features, while a sparse memory module preserves only the most useful evidence for anatomical question answering?

This framing is technically defensible, small enough to demonstrate on a MacBook, and ambitious enough to communicate a meaningful research direction.

The project should be presented as a **conceptual prototype**, not as a clinically validated diagnostic system. Its purpose is to show how representation learning, interpretability, and memory compression can be connected in a reproducible experiment.

2. The conceptual journey

Stage	Concept	Role in the story	Important distinction
1	RSNA knee abnormality research	Provides the medical problem: understanding abnormalities in knee imaging and answering anatomy-related questions	A competition or dataset is not automatically a clinical validation study
2	BioMedCLIP	Supplies a biomedical image-text embedding space for images and clinical or anatomical text	BioMedCLIP aligns image and text representations; it does not by itself provide mechanistic explanations

3	Masked autoencoders	Establish the intuition that missing visual content can be inferred from visible context	MAEs generally reconstruct pixels or patch-level visual targets
4	JEPA	Reframes masked prediction as prediction in representation space rather than exact pixel reconstruction	I-JEPA predicts target-block embeddings from a context block and is non-generative [1]
5	Sparse autoencoders	Decompose dense activations into a larger set of sparse latent features	A sparse feature can be interpretable, but sparsity alone does not guarantee a clinically meaningful concept
6	MovieChat-style sparse memory	Shows how dense streams can be compressed into a smaller memory for long-context question answering	MovieChat addresses long-video memory, not medical representation interpretability [3]
7	LLM interpretability and episodic memory	Motivates the broader question of whether useful, human-readable features can act as memory units	Claims about a “10,000×” improvement should be tied to a precise paper and metric before publication

BioMedCLIP is a particularly useful anchor because it was pretrained on PMC-15M, a large biomedical image-caption corpus, and uses biomedical adaptations including PubMedBERT and a domain-specific image encoder [2]. It is therefore a reasonable source of embeddings for a prototype, while the RSNA knee data provides a domain-specific test setting.

The key bridge is this: **BioMedCLIP produces dense representations; a sparse autoencoder can be trained over those representations; a memory policy can retain only the most relevant sparse features for a question.** JEPA then becomes a complementary training principle for learning semantic representations from partial observations, rather than a mandatory replacement for BioMedCLIP.

3. Recommended project scope

Project title

Sparse Anatomical Memory: Interpreting BioMedCLIP Representations for Knee-Imaging QA

Alternative titles include “**From Dense Medical Embeddings to Sparse Anatomical Concepts**” and “**Can Sparse Memory Make Medical Vision-Language Models More Interpretable?**”

Research question

Given a frozen biomedical vision-language encoder, can a sparse autoencoder learn latent features that are more selectively associated with anatomy, pathology, or imaging context than the original dense embedding dimensions?

Optional extension

Can the same sparse features be compressed into a question-aware memory that supports anatomical QA with less context than the original dense token sequence?

The first question should be the main deliverable. The memory and QA component should be an extension, because it introduces additional sources of error and is more difficult to evaluate convincingly on limited hardware.

4. Minimal viable experiment for a MacBook with MPS

Recommended architecture

Use a **frozen BioMedCLIP image encoder** and extract one embedding per image, or a small set of patch/token embeddings if the implementation exposes them. Train only a small sparse autoencoder on these activations.

The basic model is:

$$z = \text{ReLU}(W_e(x - b_e))$$

$$\hat{x} = W_d z + b_d$$

where x is a dense BioMedCLIP activation, z is the sparse latent representation, and $\hat{x}$ is the reconstruction.

A practical loss is:

$$\mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1$$

The first term preserves information. The second term encourages only a small number of latent features to activate. If feature collapse occurs, add a decoder-column normalization step or use a more careful sparsity schedule.

Suggested scale

Component	Small prototype setting
Input activation width	512–1024 dimensions, depending on the selected BioMedCLIP output
Sparse latent width	$2 \times -8 \times$ the input width
Training samples	1,000–10,000 images or image embeddings
Batch size	16–64, adjusted to available unified memory
Precision	Start with float32; test mixed precision only after correctness is established
Training duration	A few thousand to tens of thousands of steps
Device	<code>mps</code> when available, otherwise CPU fallback
Trainable parameters	Sparse autoencoder only; keep BioMedCLIP frozen

Do not begin by jointly training BioMedCLIP, JEPA, the sparse autoencoder, and a language model. That would make the experiment difficult to interpret and unnecessarily expensive. The first demonstration should isolate the effect of the sparse autoencoder.

Data strategy

Use only data that you are legally and ethically allowed to access. For the prototype, create a small table containing an image identifier, a study or patient-group split, available abnormality labels, and a short anatomy/pathology text description. Keep patient-level separation between training and evaluation data to avoid leakage.

If full RSNA knee MRI data are computationally or access constrained, begin with a public, de-identified knee-imaging subset or a synthetic control set. The LinkedIn post should state exactly which data were used and should not imply that results generalize to the full RSNA challenge.

Training stages

Stage A: Baseline embeddings. Load the frozen BioMedCLIP image encoder, preprocess images consistently, and save embeddings to disk. This makes repeated experiments inexpensive and allows the model to run within MacBook memory limits.

Stage B: Sparse autoencoder training. Train the SAE to reconstruct the dense embeddings while penalizing latent activation. Record reconstruction loss, average number of active features, dead-feature percentage, and activation distributions.

Stage C: Feature inspection. For each sparse feature, rank images by activation. Display the top-activating images and compare them with known labels or anatomy descriptions. This is more persuasive than showing only a loss curve.

Stage D: Text alignment. Encode a controlled vocabulary such as “ACL,” “PCL,” “meniscus,” “cartilage,” “bone marrow edema,” “effusion,” and “normal knee.” Measure whether individual SAE features correlate with image labels or with image–text similarity scores.

Stage E: Question-aware sparse memory. For a question such as “Is there evidence of meniscal abnormality?” , select the top-k sparse features whose image activations and text concept similarities are most relevant. Pass only those selected features, together with a short evidence summary, to a lightweight QA layer or a manually defined answer rule.

5. Evaluation plan

The project needs three separate evaluation axes because reconstruction, interpretability, and QA quality are different objectives.

Evaluation axis	Example metric	What it answers
Representation fidelity	Mean squared reconstruction error and cosine similarity between x and $\hat{x}$	Did the SAE preserve the original representation?
Sparsity	Mean active features per example, L0 proxy, and dead-feature rate	Is the representation actually sparse and usable?

Feature interpretability	Label correlation, precision@k of top-activating images, entropy, and human inspection	Do features correspond to recognizable anatomy or pathology?
Text alignment	Correlation between feature activation and BioMedCLIP image-text similarity	Do features connect to meaningful biomedical concepts?
QA performance	Accuracy, macro-F1, or exact-match depending on the task	Does sparse memory preserve useful evidence for questions?
Compression	Number of retained features or tokens relative to the dense representation	What is the memory reduction, and what accuracy is lost?

A compelling result would not merely be “the SAE reconstructs well.” It would show a trade-off curve: **as the number of retained sparse features decreases, interpretability and QA performance remain stable up to a useful compression point.**

Avoid claiming clinical interpretability from automated neuron labels alone. Automated labels are hypotheses. Show the top-activating examples, report failure cases, and describe the features as **candidate anatomical concepts** unless a qualified expert has reviewed them.

6. How JEPA fits without overcomplicating the prototype

JEPA should be described as a **future representation-learning extension**, not necessarily included in the first training run.

In I-JEPA, the model uses a visible context block to predict the representations of target blocks. Its goal is to learn semantic structure in embedding space rather than reconstruct every pixel [1]. For knee MRI, a future experiment could mask spatial regions or slices and ask a predictor to estimate the latent representation of the missing anatomy. The SAE could then be trained on the resulting representations to test whether JEPA-style features are more stable or more anatomically coherent than features from a standard image encoder.

The proposed progression is therefore:

Plain Text

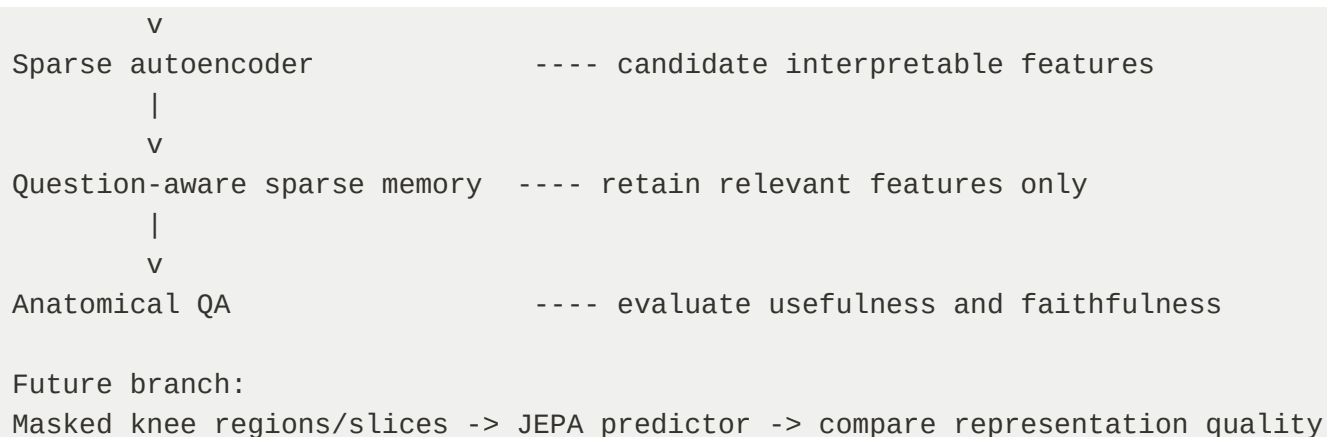
Knee images + reports

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Frozen BioMedCLIP embeddings ---- baseline dense representation

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This distinction makes the project easier to explain: **BioMedCLIP is the biomedical embedding source, SAE is the interpretability mechanism, sparse memory is the compression mechanism, and JEPA is a possible alternative or upstream representation-learning objective.**

7. How to connect MovieChat and LLM memory accurately

MovieChat is useful as an analogy for the memory problem. It addresses long-video understanding by converting a dense stream of visual information into a sparse memory and uses Transformer tokens as memory carriers; the authors report the MovieChat-1K benchmark with 1,000 long videos and 14,000 manual annotations [3].

The medical analogue is not “a knee MRI is a movie.” The more precise analogy is that a multi-slice MRI study contains repeated, highly redundant observations. A memory layer could retain the most informative slice-level, region-level, or concept-level evidence instead of passing every token to a downstream QA model.

For LLM interpretability, the safe claim is that sparse features may provide a more inspectable interface between an encoder and a reasoning system. Do not claim that sparse features automatically reveal the model’s true internal reasoning, nor that they guarantee faithful explanations. State that the experiment tests whether sparse features are **predictive, selective, and behaviorally useful**.

The phrase “10,000 times” should be removed unless you can identify the exact publication, baseline, and metric. It may refer to a 10,000-frame context, a token-compression factor, or an interpretability-related quantitative result; those are not interchangeable claims.

8. LinkedIn post structure

Opening hook

“While studying knee abnormalities in the RSNA setting, I kept encountering the same question from different directions: how can a model preserve the important evidence while making its internal representation easier to inspect?”

Paragraph 1: The starting point

Explain that biomedical image understanding is not only a classification problem. For anatomy-related QA, the system should connect visual evidence with concepts such as meniscus, ACL, cartilage, effusion, or bone marrow abnormality.

Paragraph 2: The conceptual connection

Introduce BioMedCLIP as a biomedical image-text representation model, then explain that its embeddings are dense and therefore difficult to inspect directly. Mention that MAEs reconstruct missing visual content, whereas JEPA predicts missing content in representation space.

Paragraph 3: The hypothesis

Present the hypothesis: a sparse autoencoder may transform dense biomedical embeddings into a larger dictionary of sparse candidate concepts. A question-aware memory mechanism may then retain only the features relevant to the current anatomical question.

Paragraph 4: The experiment

Describe the small MPS experiment: freeze BioMedCLIP, extract embeddings, train an SAE, rank top-activating images for each feature, align features with anatomy terms, and compare dense versus sparse QA evidence.

Paragraph 5: The evaluation

State that you will measure reconstruction fidelity, sparsity, feature-label alignment, compression, QA performance, and failure cases. Emphasize that the objective is not only higher accuracy but also a more inspectable representation.

Closing question

“Can sparse memory become a useful interface between biomedical vision-language models and anatomical reasoning—or does compression remove the very evidence needed for a faithful answer?”

Invite feedback from researchers working on medical imaging, mechanistic interpretability, JEPA, and memory-augmented models.

9. Suggested visual assets for the post

Create one clean diagram rather than a dense collage. The diagram should show the flow from knee image to BioMedCLIP embedding, then to sparse features, then to a question-aware memory and an anatomical answer. Add a small side-by-side panel showing a dense activation vector versus a sparse activation vector.

A second visual can show the top-activating examples for three candidate features. Label them conservatively as “Feature 17: candidate meniscus-related pattern,” “Feature 42: candidate fluid-related pattern,” and “Feature 88: candidate imaging-position pattern.” The word **candidate** is important until expert review confirms the interpretation.

10. Proposed repository structure

Plain Text

```
sparse-anatomical-memory/  
├── README.md  
├── requirements.txt  
├── configs/  
│   └── sae_small.yaml  
├── data/  
│   ├── metadata_template.csv  
│   └── README.md  
├── notebooks/  
│   ├── 01_extract_biomedclip_embeddings.ipynb  
│   ├── 02_train_sparse_autoencoder.ipynb  
│   └── 03_visualize_features_and_qa.ipynb  
├── src/  
│   ├── extract_embeddings.py  
│   ├── sparse_autoencoder.py  
│   ├── train_sae.py  
│   ├── evaluate_features.py  
│   └── sparse_memory.py  
├── reports/  
│   ├── figures/  
│   └── results.csv  
└── linkedin_post.md
```

The README should include the exact model checkpoint, preprocessing steps, dataset license, train/evaluation split, device configuration, and known limitations.

11. Limitations to state explicitly

The prototype will likely use a small or nonrepresentative sample, a frozen general biomedical encoder rather than a knee-specific MRI foundation model, and weak or incomplete labels. MRI is three-dimensional and often requires sequence- and slice-level reasoning, while a single pooled embedding can discard spatial information. A sparse feature that correlates with a label may encode acquisition protocol, scanner, view, or artifacts rather than pathology. Finally, a QA answer can be correct for the wrong reason, so faithfulness requires counterfactual tests or expert review.

The project should therefore be positioned as **an interpretability and representation-learning study**, not as a diagnostic tool or clinical decision-support system.

12. Recommended order of execution

First, reproduce a frozen BioMedCLIP embedding extraction pipeline on a small, permitted sample. Second, train and visualize a sparse autoencoder before adding any QA component. Third, test whether the top SAE features align with controlled anatomical terms and known labels. Fourth, add a simple top-k sparse memory policy. Only after this baseline works should you explore JEPA-style masked slice or region prediction.

The most publishable and LinkedIn-friendly claim is not “I built a medical AGI.” It is:

I built a small experiment to test whether dense biomedical representations can be converted into sparse, inspectable candidate concepts without losing the evidence needed for anatomical question answering.

References

- [1]: <https://arxiv.org/abs/2301.08243> “Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture”
- [2]: <https://arxiv.org/html/2303.00915v3> “BiomedCLIP: A Multimodal Biomedical Foundation Model Pretrained from Fifteen Million Scientific Image-Text Pairs”
- [3]: <https://www.microsoft.com/en-us/research/publication/moviechat-from-dense-token-to-sparse-memory-for-long-video-understanding/> “MovieChat: From Dense Token to Sparse Memory for Long Video Understanding”

- [4]: <https://arxiv.org/abs/2510.26411> “MedSAE: Dissecting MedCLIP Representations with Medical Sparse Autoencoders”
- [5]: <https://www.rsna.org/artificial-intelligence/ai-image-challenge/knee-mri-ai-challenge> “RSNA Knee Abnormality Detection AI Challenge”
- [6]: <https://github.com/wenhaochai/MovieChat> “MovieChat project repository”
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Short version of the project in one sentence

Freeze BioMedCLIP, train a sparse autoencoder on its knee-imaging representations, inspect the top-activating candidate anatomical features, and test whether a question-aware sparse memory can support anatomical QA with less information but more interpretability.

Terms to verify before publishing

The names “VoxSight,” “Episodic Memory,” and “LLVM” may refer to specific papers, projects, or may be shorthand or typographical variants. Before making them central to the post, add the exact paper titles, authors, links, and the precise meaning of any quantitative claim associated with them—especially the reported “10,000×” figure.

Prepared as a project and communication outline; not medical advice and not a clinical validation protocol.